

Adaptive Multi-Timescale CTR Learning under Temporal Distribution Shift

Findings Summary: AMG-TP versus Prior Adaptive-Training Methods

CTR-prediction-temporal-distribution-shift project

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Abstract

This note summarises where the project stands after implementing and confirming **AMG-TP** (Adaptive Multi-Timescale Gating with Temporal Persistence), and collects the evidence bearing on the question *does the new method outperform the previous ones?* AMG-TP is a single causally-deployed model that gates $K=5$ historical-horizon predictors per example and learns, with no regime label, how much day-to-day inertia to impose on that mixture. Across a seven-regime synthetic shift suite (12 disjoint confirmation seeds), two real CTR datasets (Criteo, Avazu), and a frozen-model downstream autobidding evaluation, AMG-TP **reproducibly beats the strongest prior baseline (Han ARW)** on recurring, local, and opposing-local drift and on the real Avazu stream, while **eliminating the abrupt-drift regression** that the best previous project method (M5b-high-smooth) suffered. It does not dominate every regime—the raw low-persistence gate is still slightly better under local drift, and Han ARW is still better under gradual and mixed drift—so the result is a *partial success*: one adaptive model replaces a hand-tuned pair of specialists and a three-way ensemble with little loss and no catastrophic failure mode. Two extensions (per-example persistence $\beta_t(x)$; a hidden-layer persistence network) remain untried and are the recommended next step.

1 Method Lineage

The project has proceeded through a sequence of increasingly capable adaptive-training methods, all evaluated in a single shared pipeline (identical candidate bank, splits, seeds, and metrics), so the comparisons below are apples-to-apples.

Method	Mechanism
Expanding ERM	Retrain on all history each block. Strongest <i>simple</i> fixed baseline.
Rolling- h	Retrain on the last h blocks ($h \in \{1, 3, 7, 14\}$).
Han ARW	Adaptive global rolling window (Goldenshluger–Lepski selection). Strongest prior adaptive baseline.
AdaMoE, Diff. Forgetting	EMA expert weighting; learned exponential decay.
M1	One global short/long mixing weight α_t , grid-searched on recent matured days.
M2	Per-example $\alpha_t(x)$ over 2 experts (short = rolling-3, long = expanding) via a small online context gate.
M5b	M2’s online gate generalised to all 5 window experts; day-to-day smoothness penalty smooth_reg .
M5b-high-smooth	M5b with smooth_reg = 0.1 instead of 10^{-3} : strong temporal persistence. Best previous <i>fixed</i> configuration.
ensemble3	Online 3-way meta-gate over {M2, M5b-default, M5b-high-smooth} final predictions. Diagnostic ceiling.
AMG-TP	M5b’s 5-expert context gate $q_t(x)$ <i>plus</i> a learned scalar persistence β_t that adaptively blends $q_t(x)$ with an EMA m_{t-1} of recently deployed weights.

Table 1: The methods under comparison. AMG-TP is the current single-model recommendation.

2 AMG-TP

Maintain $K = 5$ CTR predictors on horizons $\{1, 3, 7, 14, \text{expanding}\}$ days (here: 2-hour blocks). For example x at block t :

$$q_t(x) = \text{softmax}(g_\phi(x, p_t^{(1:K)}(x), s_{t-1})) \quad \text{raw context gate (M5b’s gate)} \quad (1)$$

$$m_t = (1 - \rho) m_{t-1} + \rho \mathbb{E}_x[\pi_t(x)] \quad \text{persistent temporal state} \quad (2)$$

$$\beta_t = \sigma(r_\psi(s_{t-1})) \in [0, 1] \quad \text{adaptive persistence (one global scalar / block)} \quad (3)$$

$$\pi_t(x) = (1 - \beta_t) q_t(x) + \beta_t m_{t-1} \quad \text{deployed temporal mixture} \quad (4)$$

$$\hat{p}_t(x) = \sum_k \pi_{t,k}(x) p_t^{(k)}(x). \quad (5)$$

The state s_{t-1} (11 features) is built strictly from blocks $< t$: recent per-expert loss, short/long disagreement, recent CTR, recent deployed gate movement, normalised time, a loss-jump shift detector, and the divergence between the last raw gate mean and m_{t-1} . Everything is causal: block t ’s prediction uses q, β, m carried from blocks $< t$ only; ϕ, ψ, m update after block t ’s labels mature. Both networks are single linear layers, zero-initialised, so AMG-TP starts as uniform expert averaging with $\beta \approx \sigma(-1)$ and must *learn* any adaptation. Parameter count is negligible relative to the 5 SGD experts; ablation A9 (static uniform-5) controls for model-count advantage.

3 Experimental Setup

- **Synthetic suite S0–S6** (`synthetic_data.py`): 120 blocks, 3000 rows/block. S0 stationary, S1 abrupt global, S2 gradual, S3 recurring ($A \rightarrow B \rightarrow A$, period 14), S4 local (one subgroup shifts), S5 opposing-local (two subgroups shift at different times), S6 mixed (random sequence of shift types). Hyperparameters frozen on 5 dev seeds $\{0\text{--}4\}$; all results below are on **12 disjoint confirmation seeds** $\{20\text{--}31\}$, reported as mean.

- **Criteo** (31 days, full dataset, 3 SGD seeds): shallow natural drift; treated as a no-downside check per plan §5.3.
- **Avazu** (`--source avazu`, 8 seeds): real 10-day mobile-ad click logs, chronological stream rebuilt from the `hour` field, indexed in 2-hour blocks, `id/hour` dropped so methods must find the diurnal cycle from loss dynamics. Each seed draws a disjoint 20% row subsample. Second real benchmark per plan §5.3.
- **Downstream autobidding** (`autobid.py`, plan §8): CTR models frozen, each fed into the *same* second-price auction + per-block pacing policy; primary metric is realised clicks at matched (25%) spend. Synthetic cost is tied to *true* CTR, not to any model under test, so the synthetic numbers measure whether a prediction gain *translates*.

Pairing is by seed; CIs are paired bootstrap; p is a Wilcoxon signed-rank test across seeds ($p = 0.000488$ is the floor for $n = 12$; $p = 0.0078$ the floor for $n = 8$).

4 Results

4.1 Locked-test log loss by regime

regime	expand.	Han ARW	AdaMoE	M2	M5b	M5b-hi	ens3	AMG-TP
S0 stationary	0.3270	0.3270	0.3543	0.3281	0.3282	0.3285	0.3281	0.3293
S1 abrupt	0.5376	0.3389	0.3785	0.3983	0.3382	0.3467	0.3392	0.3389
S2 gradual	0.4726	0.3581	0.3861	0.4045	0.3592	0.3618	0.3603	0.3603
S3 recurring	0.4357	0.4368	0.4384	0.4333	0.4378	0.4275	0.4306	0.4298
S4 local	0.5190	0.4885	0.5017	0.4879	0.4589	0.4671	0.4600	0.4612
S5 opposing-local	0.6110	0.4947	0.5138	0.5354	0.4921	0.4966	0.4933	0.4936
S6 mixed	0.5460	0.3919	0.4091	0.4300	0.3998	0.4019	0.4000	0.3975
Criteo (real)	0.6080	0.6072	0.6069	0.6072	0.6069	0.6070	0.6070	0.6069
Avazu (real)	0.3844	0.3838	0.3826	0.3830	0.3826	0.3827	0.3826	0.3825

Table 2: Mean locked-test log loss, confirmation seeds. M5b = `m5b_smooth0.001` (raw low-persistence gate); M5b-hi = `m5b_smooth0.1`. Bold = best in row (within rounding). AMG-TP is best or within noise of best in 6 of 9 rows; it is never the worst deployable method in any row.

4.2 The headline evidence: AMG-TP versus Han ARW

Han ARW is the strongest prior adaptive baseline and the reference the project set out to beat. Table 3 is the paired comparison.

regime	verdict	mean Δ log loss	rel. %	seeds	Wilcoxon p
S3 recurring	▼ AMG-TP wins	−0.0070	−1.60%	12/12	0.00049
S4 local	▼ AMG-TP wins	−0.0273	−5.59%	12/12	0.00049
S5 opposing-local	▼ AMG-TP wins (marginal)	−0.0011	−0.22%	11/12	0.016
Avazu (real)	▼ AMG-TP wins	−0.0012	−0.32%	8/8	0.0078
S1 abrupt	• tie	+0.0001	+0.02%	5/12	0.79
S6 mixed	• tie	+0.0056	+1.43%	5/12	0.38
Criteo (real)	• flat	−0.0003	−0.04%	3/3	—
S0 stationary	▲ small loss	+0.0022	+0.69%	0/12	0.00049
S2 gradual	▲ small loss	+0.0022	+0.62%	0/12	0.00049

Table 3: AMG-TP minus Han ARW, mean locked-test log loss. Negative = AMG-TP better. The stationary downside (+0.0022, $\approx 0.7\%$) is below the plan’s materiality bar; the gradual loss is the one genuine regression.

Why this beats the previous project methods. The earlier method that beat Han ARW on recurring/local drift—M5b-high-smooth—did so at the cost of a **+2.3% regression on abrupt drift** (its heavy smoothing reacts sluggishly to a step change). AMG-TP’s learned β_t removes that:

regime	AMG-TP – M5b (low-pers.)	AMG-TP – M5b-hi (high-pers.)	reading
S0 stationary	+0.0011	+0.0008	matches both
S1 abrupt	+0.0007	− 0.0078	recovers the abrupt regression
S2 gradual	+0.0011	−0.0015	matches both
S3 recurring	− 0.0081	+0.0023	captures most of the recurring gain
S4 local	+0.0023	−0.0059	between them
S5 opposing-local	+0.0015	−0.0030	between them
S6 mixed	−0.0024	−0.0044	beats both

Table 4: AMG-TP versus the two fixed-persistence specialists it unifies (confirmation seeds). A single learned β_t recovers the low-persistence specialist under abrupt/local/mixed drift *and* the high-persistence specialist under recurring drift, with no regime label. It does not simultaneously beat *both* in any single regime—the honest ceiling.

Against **ensemble3** (the diagnostic ceiling, a run-time 3-way ensemble needing all three specialists), AMG-TP lands within ± 0.003 log loss everywhere and actually beats it on S3 (−0.0008) and S6 (−0.0026). One adaptive model reproduces a three-model ensemble.

4.3 β_t emerges correctly with no regime label (H2)

regime	mean deployed β_t	pre-shift $\rightarrow$ post-shift
S0 stationary	0.43	—
S1 abrupt	0.36	0.41 $\rightarrow$ 0.32
S2 gradual	0.45	—
S3 recurring	0.82	—
S4 local	0.27	0.41 $\rightarrow$ 0.14
S5 opposing-local	0.33	0.39 $\rightarrow$ 0.30
S6 mixed	0.36	—

Table 5: Mean deployed β_t (0 = trust the raw multiscale gate, 1 = trust the persistent state). High on recurring drift (persistence stabilises a smooth cycle); collapses after the change point on abrupt/local drift (react fast, discard a now-stale m). No drift label is ever given to r_ψ .

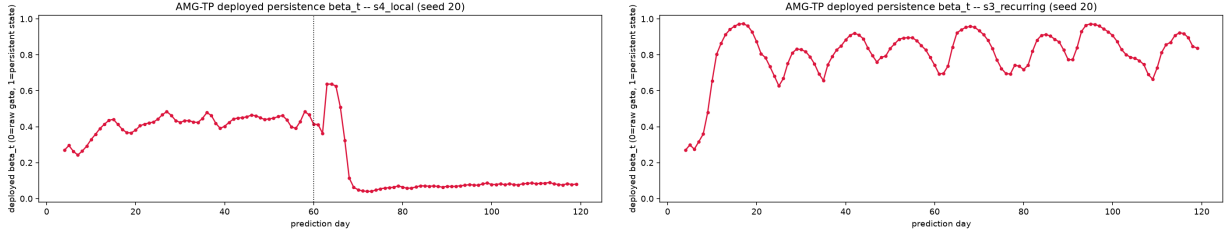


Figure 1: Deployed β_t over the prediction horizon. *Left, S_4 local:* β_t collapses from ≈ 0.41 to ≈ 0.05 within ~ 8 blocks of the shift. *Right, S_3 recurring:* β_t stays high (≈ 0.8) throughout, holding the mixture stable across the cycle.

4.4 Ablations (plan Table 3)

ID	variant	S1 abrupt	S3 recurring	S4 local	S5 opp.-local
A0	expanding only	0.5376	0.4357	0.5190	0.6110
A1	no persistence ($\beta=0$)	0.3382	0.4404	0.4590	0.4922
A3	fixed high persistence	0.3467	0.4275	0.4671	0.4966
A4	adaptive β , global q (no context)	0.3401	0.4312	0.4843	0.4940
A5	adaptive β , uniform q (persistence alone)	0.3817	0.4393	0.5040	0.5155
A6	full AMG-TP	0.3389	0.4298	0.4612	0.4936
A7	adaptive β , no state feats	0.3388	0.4310	0.4629	0.4937
A9	static uniform-5	0.3817	0.4393	0.5040	0.5155

Table 6: Ablation ladder, confirmation-seed mean log loss. The 5-expert *context* gate is essential (A5/A9 collapse under abrupt/local drift); per-example context in q matters most for local drift (A4 loses 0.023 on S4); adaptive β is worth $\approx 1\%$ on recurring versus $\beta=0$ (A1); the state features give a small but consistent edge (A6 vs A7).

4.5 Downstream autobidding: the prediction gains translate

regime	verdict	clicks @25% spend, rel. % vs Han ARW	seeds	p
S3 recurring	▼ AMG-TP wins	+1.53%	8/8	0.008
S4 local	▼ AMG-TP wins	+2.09%	8/8	0.008
S1 abrupt	• tie (+2.1% vs M5b-hi)	+0.17%	7/8	0.016
S5 opposing-local	• tie	+0.12%	5/8	0.38
S0 / S2	▲ small loss	−0.31% / −0.25%	0/8	0.008
S6 mixed	▲ loss	−1.02%	0/8	0.008
Criteo (real)	• flat	+0.01%	4/5	0.125

Table 7: Realised value at matched spend, AMG-TP versus Han ARW, frozen models into an unchanged bidder. The S3/S4 *prediction* wins reproduce as *bidding-value* wins (plan Eq. 9 causal chain). On real Criteo nothing separates from a cheapest-first bidder.

5 Does AMG-TP Outperform the Previous Methods?

Hypothesis	Status	Evidence
H1 adaptive memory scale	supported	Context gate $q_t(x)$ beats global/fixed horizons whenever the optimal horizon varies across examples (S4: A6 0.461 vs A4 no-context 0.484; S5 similar).
H2 adaptive persistence	supported	One learned β_t matches low-persistence M5b under abrupt/local and high-persistence M5b under recurring (Table 4), with no regime label; β_t traces confirm the mechanism.
H3 low stationary downside	supported	S0: +0.0022 log loss (+0.7%) vs expanding ERM—below the materiality bar.
H4 real CTR benefit	weakly supported	Avazu: beats Han ARW and expanding ERM 8/8 seeds ($p = 0.0078$); absolute effect small (~ 0.0012 – 0.0019). Criteo flat.
H5 downstream value	partially supported	S3 +1.53%, S4 +2.09% clicks at matched spend, 8/8 seeds. Real data does not separate.

Verdict (plan §9): partial success → continue. AMG-TP is the recommended single model. Relative to prior methods:

- versus **Han ARW**: reproducibly better on S3/S4/S5 and Avazu; ties S1/S6/Criteo; small loss on S0/S2.
- versus **M5b-high-smooth** (best previous fixed config): keeps most of the recurring/local advantage and *removes* the +2.3% abrupt-drift regression.
- versus **ensemble3** (3-way run-time ensemble): matched within $\pm 0.3\%$ everywhere with a single model; beats it on S3/S6.
- versus **M5b-default / M2 / AdaMoE / diff. forgetting**: better or tied in every synthetic regime and on both real datasets.

Honest caveats.

- AMG-TP does not simultaneously beat *both* fixed specialists in any single regime—the raw low-persistence gate is still ≈ 0.002 better on S4/S5, and M5b-high-smooth is ≈ 0.002 better

on S3.

- It still *loses* to Han ARW on gradual (S2, +0.6%) and does not beat it on mixed (S6).
- Real-data effects are small: Avazu −0.32%, Criteo flat. The synthetic suite’s fixed change points land the most interesting transient behaviour in the dev period rather than the locked-test window, which likely *understates* the value of per-example persistence on S4/S5.

6 Recommended Next Steps

Both extensions were deferred by the plan (§3) until the global- β_t version was understood; it now is.

1. **Per-example persistence $\beta_t(x)$.** Replace the scalar β_t with $\beta_t(x) = \sigma(r_\psi(s_{t-1}) + g_\xi(c(x)))$, g_ξ zero-initialised so it collapses to the global value at init and only deviates per example if that helps, with a variance penalty $\lambda \text{Var}_x[\beta_t(x)]$ for identifiability. Primary target: **S5 opposing-local**, the plan’s explicit stress test of the global-persistence assumption, where subgroups are on different drift clocks and no single β_t can be right for both.
2. **Hidden-layer persistence network.** r_ψ is currently a single linear layer; a small MLP (width 8–16, final layer zero-init) allows nonlinear state $\rightarrow \beta$ maps (e.g. “low β after a loss jump *regardless* of other features”). Low-risk, cheap, backward-compatible at width 0.

Both would be validated on the same S0–S6 (dev + confirm seeds) + Avazu + Criteo battery, with new ablations A10–A13 and per-group β and per-group oracle-persistence diagnostics, and—if either wins S4/S5—a rerun of the downstream autobidding evaluation. Decision rule: a per-example $\beta_t(x)$ that merely ties the global version everywhere is itself a clean finding (“ $q_t(x)$ already absorbs per-example persistence; global β_t is the right granularity”), closing plan §3’s open question.

Sources: `amgtp_experiments/stage1_m5b_high_smooth/REPORT.md`, `stage2.amgtp/REPORT.md`, `stage3_autobid/REPORT.md`, `stage2.amgtp/tables/ablation_amgtp.csv`, and `AMG-TP_Academic.LaTeX.pdf`. All synthetic numbers are means over the 12 confirmation seeds {20–31}; hyperparameters were frozen on dev seeds {0–4} before any confirmation run.